RAG Learning

4.6 Vectors Higher

OCR GCSE Mathematics J560 — Component 4 (Geometry and Measures) — Papers 2, 3 (Higher only)

4.6.1 Vector Notation

A **vector** has both **magnitude** (size) and **direction**. A scalar has magnitude only. Vectors are used to represent displacements, velocities, and forces.

Ways to Write a Vector

Notation	Meaning	Example
$\mathbf{a}$ (bold)	Vector named a (used in printed text)	$\mathbf{a} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$
$\underline{a}$ (underlined)	Handwritten form of bold notation	$\underline{a}$
$\overrightarrow{AB}$	Vector from point A to point B	$\overrightarrow{AB} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$
$\begin{pmatrix} x \\ y \end{pmatrix}$	Column vector: x right, y up	$\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ means 2 left, 5 up

Magnitude (modulus) of a vector:

If $\mathbf{a} = \begin{pmatrix} a_x \\ a_y \end{pmatrix}$, then $|\mathbf{a}| = \sqrt{a_x^2 + a_y^2}$

Example: $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, then $|\mathbf{a}| = \sqrt{9 + 16} = \sqrt{25} = 5$

Worked Example: Find the magnitude of $\overrightarrow{AB}$ where A = (2, 5) and B = (8, 1).

$$\overrightarrow{AB} = B - A = \begin{pmatrix} 8-2 \\ 1-5 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{6^2 + (-4)^2} = \sqrt{36+16} = \sqrt{52} = 2\sqrt{13} \approx 7.21$$

Exam Tip: In handwritten work, always underline vector letters (e.g. $\underline{a}$) or use arrow notation (e.g. $\overrightarrow{AB}$). Writing just "a" (not bold or underlined) suggests a scalar, which loses marks. Notice that $\overrightarrow{BA} = -\overrightarrow{AB}$.

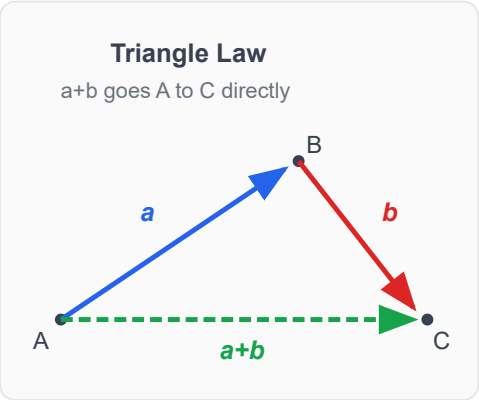
4.6.2 Adding and Subtracting Vectors

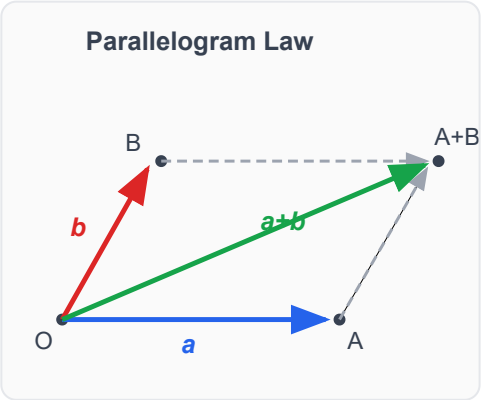
Vector addition: $\mathbf{a} + \mathbf{b}$ means "travel along $\mathbf{a}$, then travel along $\mathbf{b}$." The result is the same as going directly from start to finish.

Vector subtraction: $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$, i.e. add the negative of $\mathbf{b}$.

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix} \qquad \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}$$

Triangle Law and Parallelogram Law





Worked Example: $\mathbf{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$. Calculate $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$.

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 4 + (-2) \\ 1 + 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

$$\mathbf{a} - \mathbf{b} = \begin{pmatrix} 4 - (-2) \\ 1 - 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

4.6.3 Scalar Multiplication

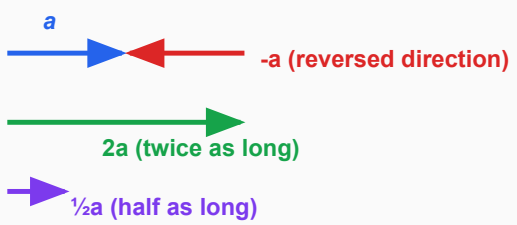
Multiplying a vector by a **scalar** (a number) changes its magnitude but not its direction (unless the scalar is negative, which reverses direction).

$$k\mathbf{a} = k \begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} ka_x \\ ka_y \end{pmatrix}$$

Parallel vectors: Two vectors are parallel if one is a scalar multiple of the other. If $\mathbf{b} = k\mathbf{a}$ for some scalar k, then $\mathbf{a} \parallel \mathbf{b}$.

Negative vector: $-\mathbf{a}$ has the same magnitude as $\mathbf{a}$ but points in the opposite direction.

All vectors above are parallel to each other



Worked Example: $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$. Find $3\mathbf{a}$, $-\mathbf{a}$, and $\frac{1}{2}\mathbf{a}$.

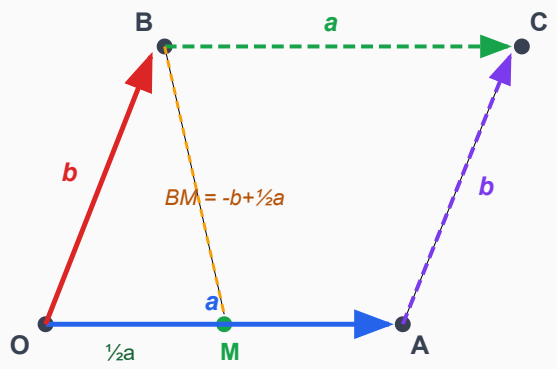
$3\mathbf{a} = \begin{pmatrix} 6 \\ -9 \end{pmatrix} \quad -\mathbf{a} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \quad \frac{1}{2}\mathbf{a} = \begin{pmatrix} 1 \\ -1.5 \end{pmatrix}$

4.6.4 Finding Vectors from Diagrams

Given a diagram where $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, you can express any other vector in terms of $\mathbf{a}$ and $\mathbf{b}$ by finding a path between the two points using vectors you already know.

Key strategy: To find $\overrightarrow{XY}$, write it as $\overrightarrow{XO} + \overrightarrow{OY}$ (go via O or any convenient intermediate point). Remember $\overrightarrow{XO} = -\overrightarrow{OX}$.

OABC is a parallelogram. M is midpoint of OA.



Worked Example: OABC is a parallelogram. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, M is the midpoint of OA.

(a) Find $\overrightarrow{AB}$

$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$

(b) Find $\overrightarrow{BM}$

$\overrightarrow{BM} = \overrightarrow{BO} + \overrightarrow{OM} = -\mathbf{b} + \frac{1}{2}\mathbf{a}$

(c) Find $\overrightarrow{BC}$ (note BC is parallel to OA, so $\overrightarrow{BC} = \mathbf{a}$)
$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = \mathbf{b} + \mathbf{a} = \mathbf{a} + \mathbf{b}$

Exam Tip: Always trace a path between the two points using vectors with known values. Keep track of the direction: going against an arrow means negating the vector. Show all intermediate steps.

4.6.5 Vector Proofs

Vector proofs typically ask you to show that:

- **Two lines are parallel**: show one vector is a scalar multiple of the other
- **Three points are collinear**: show the vector between any two of the points is parallel to the vector between another pair (and they share a point)
- **A point is a midpoint**: show the vector from each end to the point is equal in magnitude (half of the segment's vector)

Proving lines are parallel: If $\overrightarrow{XY} = k\overrightarrow{PQ}$ for a non-zero scalar k, then XY is parallel to PQ.

Proving points are collinear: Show $\overrightarrow{AB} = k\overrightarrow{AC}$ (same direction, share point A), so A, B, C are collinear.

Worked Example: Proof: midpoint and parallel line
OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$. M is the midpoint of OA, N is the midpoint of OB. Prove that MN is parallel to AB and $MN = \frac{1}{2}AB$.

$\overrightarrow{OM} = \frac{1}{2}\mathbf{a}, \quad \overrightarrow{ON} = \frac{1}{2}\mathbf{b}$

$\overrightarrow{MN} = \overrightarrow{MO} + \overrightarrow{ON} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} = \frac{1}{2}(\mathbf{b} - \mathbf{a})$

$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$

Since $\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB}$:
(1) MN is parallel to AB (scalar multiple relationship)
(2) $|MN| = \frac{1}{2}|AB|$ (the scalar is $\frac{1}{2}$) **QED**

Worked Example: Proving collinearity: $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$. Point P lies on AB such that $AP : PB = 2 : 1$. Show that O, A, P are NOT collinear but find $\overrightarrow{OP}$.

$\overrightarrow{AP} = \frac{2}{3}\overrightarrow{AB} = \frac{2}{3}(-\mathbf{a} + \mathbf{b}) = -\frac{2}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$

$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \mathbf{a} - \frac{2}{3}\mathbf{a} + \frac{2}{3}\mathbf{b} = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$

Since $\overrightarrow{OP}$ contains both $\mathbf{a}$ and $\mathbf{b}$ (which are not parallel), O, A, P are not collinear.

Worked Example: Classic exam proof: In triangle OAB, $\overrightarrow{OA} = 6\mathbf{a}$, $\overrightarrow{OB} = 6\mathbf{b}$. X is the midpoint of AB. Y is on OB such that $OY : YB = 1 : 2$. Show that A, X, Y are collinear.

$\overrightarrow{OX} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = 6\mathbf{a} + \frac{1}{2}(-6\mathbf{a} + 6\mathbf{b}) = 3\mathbf{a} + 3\mathbf{b}$

$\overrightarrow{OY} = \frac{1}{3} \times 6\mathbf{b} = 2\mathbf{b}$

$\overrightarrow{AX} = \overrightarrow{AO} + \overrightarrow{OX} = -6\mathbf{a} + 3\mathbf{a} + 3\mathbf{b} = -3\mathbf{a} + 3\mathbf{b}$

$\overrightarrow{AY} = \overrightarrow{AO} + \overrightarrow{OY} = -6\mathbf{a} + 2\mathbf{b}$

Check: is $\overrightarrow{AY}$ a multiple of $\overrightarrow{AX}$?
$\frac{2}{3}\overrightarrow{AY} = \frac{2}{3}(-6\mathbf{a} + 2\mathbf{b}) = -4\mathbf{a} + \frac{4}{3}\mathbf{b}$ — this is NOT a multiple of $\overrightarrow{AX} = -3\mathbf{a} + 3\mathbf{b}$...
Let's check $\overrightarrow{XY}$: $\overrightarrow{XY} = \overrightarrow{XO} + \overrightarrow{OY} = -(3\mathbf{a} + 3\mathbf{b}) + 2\mathbf{b} = -3\mathbf{a} - \mathbf{b}$
Is $\overrightarrow{XY} = k\overrightarrow{AX}$? $-3\mathbf{a} - \mathbf{b} = k(-3\mathbf{a} + 3\mathbf{b})$ requires $k = -1$ (from $\mathbf{a}$) and $k = -\frac{1}{3}$ (from

$\mathbf{b}$) — not equal, so A, X, Y are NOT collinear in this case. (This demonstrates the method; adjust ratios in your exam question carefully.)

Exam Tip: For collinearity proofs: (1) find $\overrightarrow{AB}$ and $\overrightarrow{AC}$, (2) show $\overrightarrow{AC} = k\overrightarrow{AB}$, (3) state "since they are parallel and share point A, A, B, C are collinear." All three steps are needed for full marks. A ratio alone without the shared point statement is incomplete.

Summary of Key Techniques

Task	Method
Find $\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\overrightarrow{OA} + \overrightarrow{OB} = \mathbf{b} - \mathbf{a}$
Find midpoint M of AB	$\overrightarrow{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$
Point dividing AB in ratio $m : n$	$\overrightarrow{OP} = \mathbf{a} + \frac{m}{m+n}(\mathbf{b} - \mathbf{a})$
Prove parallel	Show $\overrightarrow{XY} = k\overrightarrow{PQ}$, state "scalar multiple, therefore parallel"
Prove collinear	Show vectors between pairs of points are parallel AND share a common point
Magnitude of $\mathbf{a} = \begin{pmatrix} p \\ q \end{pmatrix}$	$\lvert\mathbf{a}\rvert = \sqrt{p^2 + q^2}$

4.6.6 Vectors in Three Dimensions A-Level

In 3D, a vector has three components — one for each axis (x, y, z):

$$\mathbf{a} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} = a_x\,\mathbf{i} + a_y\,\mathbf{j} + a_z\,\mathbf{k}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are unit vectors along the x-, y-, z-axes respectively.

Magnitude: $|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

All 2D vector rules (addition, subtraction, scalar multiplication, magnitude) extend directly to 3D — just include the third component throughout.

Worked Example: $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}$.

Find (a) $\mathbf{a} + \mathbf{b}$, (b) $2\mathbf{a} - \mathbf{b}$, (c) $|\mathbf{a}|$.

(a) $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$

(b) $2\mathbf{a} - \mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ -5 \\ 7 \end{pmatrix}$

(c) $|\mathbf{a}| = \sqrt{1 + 4 + 9} = \sqrt{14}$

Unit Vectors

A **unit vector** has magnitude 1. The unit vector in the direction of $\mathbf{a}$ is:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Worked Example: Find the unit vector in the direction of $\mathbf{v} = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$.

$$|\mathbf{v}| = \sqrt{9 + 0 + 16} = 5$$

$$\hat{\mathbf{v}} = \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0 \\ 0.8 \end{pmatrix}$$

Check: $|\hat{\mathbf{v}}| = \sqrt{0.36 + 0 + 0.64} = \sqrt{1} = 1$ ✓

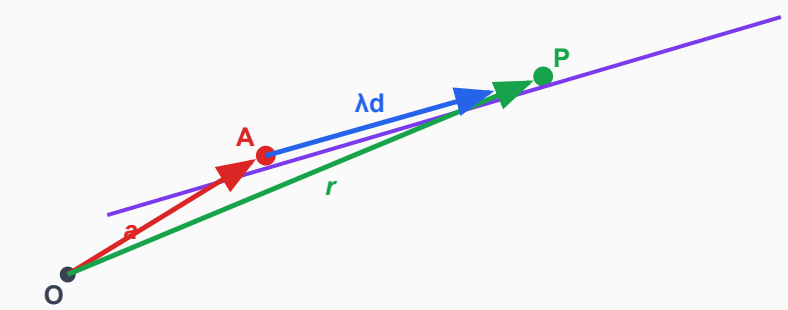
4.6.7 Vector Equation of a Line A-Level

The **vector equation of a straight line** passing through point A (with position vector $\mathbf{a}$) in the direction of vector $\mathbf{d}$ is:

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{d} \quad (\lambda \in \mathbb{R})$$

where λ is a scalar parameter. Different values of λ give different points on the line.

The line can also be written as $\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$ if it passes through points A and B, with direction vector $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$.



r = a + λd (line through A, direction d)

Worked Example: Write the vector equation of the line passing through $A(1, 2, -1)$ and $B(3, -1, 4)$.

Direction vector: $\mathbf{d} = \overrightarrow{AB} = \begin{pmatrix} 3-1 \\ -1-2 \\ 4-(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

Equation: $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

(Could equally use B as the point: $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$ — both equations represent the same line.)

Checking if a Point Lies on a Line

Substitute the coordinates of the point into the line equation and check whether the same value of λ works for all three components.

Worked Example: Does the point $P(7, -7, 14)$ lie on the line $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$?

From x: $1 + 2\lambda = 7 \Rightarrow \lambda = 3$
From y: $2 - 3(3) = 2 - 9 = -7$ ✓
From z: $-1 + 5(3) = 14$ ✓

All three components agree with $\lambda = 3$, so P lies on the line.

4.6.8 Intersecting, Parallel and Skew Lines A-Level

Two lines in 3D can be:

- **Parallel**: direction vectors are scalar multiples of each other; lines never meet.
- **Intersecting**: lines meet at exactly one point; direction vectors are not parallel.
- **Skew**: lines are not parallel and do not intersect (only possible in 3D); they lie in different planes.

Testing for Intersection

Use different parameters (λ and μ) for each line. Set the equations equal component by component to get three equations in two unknowns. If a consistent solution exists, the lines intersect.

Worked Example: Line l_1: $\mathbf{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\-1\end{pmatrix}$ Line l_2: $\mathbf{r} = \begin{pmatrix}3\\4\\0\end{pmatrix} + \mu\begin{pmatrix}2\\1\\1\end{pmatrix}$

Setting equal, component by component:
x: $1 + \lambda = 3 + 2\mu \quad \Rightarrow \quad \lambda - 2\mu = 2$ ...(i)
y: $2 + \lambda = 4 + \mu \quad \Rightarrow \quad \lambda - \mu = 2$ ...(ii)
z: $3 - \lambda = 0 + \mu \quad \Rightarrow \quad \lambda + \mu = 3$ ...(iii)

From (ii): $\lambda = \mu + 2$. Substitute into (iii): $\mu + 2 + \mu = 3 \Rightarrow \mu = \frac{1}{2}$, $\lambda = \frac{5}{2}$.

Check in (i): $\frac{5}{2} - 2 \times \frac{1}{2} = \frac{5}{2} - 1 = \frac{3}{2} \neq 2$. **Inconsistent!**

The lines are neither parallel (direction vectors $\begin{pmatrix}1\\1\\-1\end{pmatrix}$ and $\begin{pmatrix}2\\1\\1\end{pmatrix}$ are not scalar multiples) nor intersecting — they are **skew**.

Worked Example: Find the point of intersection of $l_1 : \mathbf{r} = \begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\2\\1\end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix}1\\3\\3\end{pmatrix} + \mu\begin{pmatrix}2\\1\\-1\end{pmatrix}$.

x: $\lambda = 1 + 2\mu$ ...(i)
y: $1 + 2\lambda = 3 + \mu$ ...(ii)
z: $2 + \lambda = 3 - \mu \quad \Rightarrow \quad \lambda + \mu = 1$ ...(iii)

From (i) into (iii): $(1 + 2\mu) + \mu = 1 \Rightarrow 3\mu = 0 \Rightarrow \mu = 0, \lambda = 1.$

Check in (ii): $1 + 2(1) = 3$ and $3 + 0 = 3$ ✓

Intersection point: $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$ ✓

Exam Tip: When testing for intersection: (1) use λ for line 1 and μ for line 2, (2) write three equations from the three components, (3) solve any two equations for λ and μ, (4) *check* these values satisfy the third equation. If the third equation fails, the lines are skew. Always state the conclusion clearly.

4.6.9 Angle Between Two Lines — The Scalar (Dot) Product A-Level

The **scalar product** (dot product) of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Geometrically: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$, where θ is the angle between the vectors.

Angle between two vectors:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Perpendicular vectors: $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$

Angle between two lines: find the angle between their direction vectors. If the angle $\theta > 90^\circ$, take the acute angle $180^\circ - \theta$ (since lines have no fixed orientation).

Worked Example: Find the acute angle between the lines with direction vectors $\mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$.

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 1(2) + 2(1) + 2(-2) = 2 + 2 - 4 = 0$$

Since the dot product is 0, the lines are **perpendicular**. The angle between them is 90°.

Worked Example: Find the acute angle between lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

$$\mathbf{d}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \mathbf{d}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 2 + 2 + 3 = 7$$

$$|\mathbf{d}_1| = \sqrt{4 + 1 + 1} = \sqrt{6}, \quad |\mathbf{d}_2| = \sqrt{1 + 4 + 9} = \sqrt{14}$$

$$\cos\theta = \frac{7}{\sqrt{6}\sqrt{14}} = \frac{7}{\sqrt{84}} = \frac{7}{2\sqrt{21}}$$

$$\theta = \cos^{-1}\left(\frac{7}{2\sqrt{21}}\right) \approx 40.2^\circ$$

Exam Tip: Use $|\cos\theta|$ when finding the angle between two *lines* (to always get the acute angle between them). If you use $\cos\theta$ directly and get an obtuse angle, subtract from 180°.

Complete Summary of Key Techniques

Task	Method / Formula
3D magnitude	$\lvert\mathbf{a}\rvert = \sqrt{a_x^2 + a_y^2 + a_z^2}$
Unit vector	$\hat{\mathbf{a}} = \mathbf{a}/\lvert\mathbf{a}\rvert$
Vector equation of line	$\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$
Point on a line?	Solve for λ from one component; check it satisfies the others
Parallel lines	Direction vectors are scalar multiples
Intersecting lines	Equate components (different params), solve 2 equations, verify in 3rd
Skew lines	Not parallel AND equations are inconsistent
Dot product	$\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$
Angle between vectors/lines	$\cos\theta = \dfrac{\lvert\mathbf{a}\cdot\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ (use modulus for lines)
Perpendicular vectors	$\mathbf{a}\cdot\mathbf{b} = 0$

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